WORKED SOLUTION

STEP II 2009, Question 2

Pure • Curve Sketching and Approximation

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s2-q2

The key idea

Because $a > 1$, the function a^u is strictly increasing in u, so y is largest exactly where $\sin(\pi e^x)$ is largest and smallest where it is smallest. Everything about the curve therefore comes from the inner function πe^x, which increases steadily and runs through the peaks and troughs of the sine at points that crowd closer and closer together as x grows. For the area, the trapezia have equal heights at their ends, which makes the sum collapse to a single difference of logarithms.

Solution

Part (i)

Taking logarithms, $\ln y = \ln a \, \sin(\pi e^x)$. Differentiating implicitly,

$$\frac{1}{y}\frac{dy}{dx} = \pi \ln a \, e^x \cos(\pi e^x), \qquad \text{so} \qquad \frac{dy}{dx} = \pi \ln a \, e^x \cos(\pi e^x)\, a^{\sin(\pi e^x)}.$$

Here $\ln a > 0$ (since $a > 1$), $e^x > 0$ and $a^{\sin(\pi e^x)} > 0$, so the derivative vanishes exactly when $\cos(\pi e^x) = 0$, that is, when πe^x is an odd multiple of $\frac{\pi}{2}$. Since $\pi e^x > 0$, only the positive odd multiples are possible:

$$e^x = n + \tfrac{1}{2}, \qquad x = \ln(n + \tfrac{1}{2}), \qquad n = 0, 1, 2, \ldots$$

At these points $\sin(\pi e^x) = \sin((n + \frac{1}{2})\pi) = (-1)^n$, so $y = a^{(-1)^n}$. The stationary points are

$$\left(\ln(n + \tfrac{1}{2}),\ a\right) \ (n \text{ even}), \qquad \left(\ln(n + \tfrac{1}{2}),\ \tfrac{1}{a}\right) \ (n \text{ odd}).$$

The first set are maxima and the second minima. As a^u is increasing, $y \le a^1 = a$ and $y \ge a^{-1}$ for every x, so points where $y = a$ are global maxima and points where $y = 1/a$ are global minima. Explicitly, the maxima are at $x = \ln \frac{1}{2}$, $\ln \frac{5}{2}$, $\ln \frac{9}{2}, \ldots$ and the minima at $x = \ln \frac{3}{2}$, $\ln \frac{7}{2}$, $\ln \frac{11}{2}, \ldots$

Part (ii)

For small x, $e^x \approx 1 + x$, so

$$\sin(\pi e^x) \approx \sin(\pi + \pi x) = -\sin(\pi x) \approx -\pi x,$$

using $\sin(\pi + \theta) = -\sin\theta$ and then $\sin t \approx t$ with $t = \pi x$ small. Hence

$$y \approx a^{-\pi x} = e^{-\pi x \ln a} \approx 1 - \pi x \ln a,$$

where the last step uses $e^t \approx 1 + t$ again, now with the small quantity $t = -\pi x \ln a$. This is the required approximation: the curve crosses the line $y = 1$ at $x = 0$ with gradient $-\pi \ln a$.

Ask yourself Why can't we put $\sin t \approx t$ straight into $\sin(\pi e^x)$?

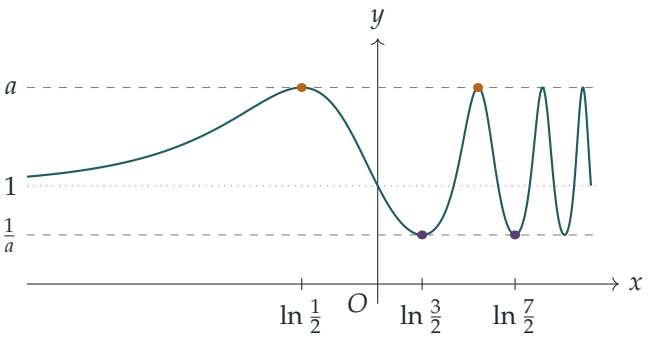
Because πe^x is close to π, not to 0, when x is small. We first have to use $\sin(\pi + \theta) = -\sin\theta$ to move the small quantity πx into the argument of the sine.

Part (iii)

Some features to put on the sketch:

- As $x \to -\infty$, $e^x \to 0$, so $\sin(\pi e^x) \to 0$ and $y \to 1$ from above (for $x < \ln\frac{1}{2}$ we have $0 < \pi e^x < \frac{\pi}{2}$, so the sine is positive and $y > 1$). The line $y = 1$ is an asymptote on the left.
- The curve rises to its first maximum a at $x = \ln\frac{1}{2}$, then oscillates for ever between a and $1/a$, always staying above the x-axis.
- It crosses $y = 1$ wherever e^x is a positive integer, at $x = 0,\ \ln 2,\ \ln 3, \ldots$, and at $x = 0$ the gradient is $-\pi \ln a$, by part (ii).
- The gap between consecutive stationary points is $\ln\frac{n+3/2}{n+1/2} \to 0$, so the oscillations get faster and faster as x increases.

The sketch shows the case $a = 2$.



Part (iv)

Count the maxima from the left: the kth maximum corresponds to $n = 2(k-1)$, so it is at $\mathrm{e}^x = 2k - \frac{3}{2}$. The $(k+1)$th maximum is at $\mathrm{e}^x = 2k + \frac{1}{2}$, and the single minimum between them is at $\mathrm{e}^x = 2k - \frac{1}{2}$. Write

$$x_1 = \ln\left(2k - \tfrac{3}{2}\right), \qquad x_2 = \ln\left(2k - \tfrac{1}{2}\right), \qquad x_3 = \ln\left(2k + \tfrac{1}{2}\right).$$

Joining (x_1, a) to $(x_2, \frac{1}{a})$ and $(x_2, \frac{1}{a})$ to (x_3, a) by straight lines, the region down to the x-axis splits into two trapezia, each with parallel sides a and $\frac{1}{a}$. So the approximate area is

$$\begin{aligned}\tfrac{1}{2}\left(a + \tfrac{1}{a}\right)(x_2 - x_1) + \tfrac{1}{2}\left(a + \tfrac{1}{a}\right)(x_3 - x_2) &= \frac{a^2+1}{2a}\,(x_3 - x_1) \\ &= \frac{a^2+1}{2a}\,\ln\frac{2k+\frac{1}{2}}{2k-\frac{3}{2}} = \frac{a^2+1}{2a}\,\ln\frac{k+\frac{1}{4}}{k-\frac{3}{4}}.\end{aligned}$$

Finally $\dfrac{k+\frac{1}{4}}{k-\frac{3}{4}} = 1 + \dfrac{1}{k-\frac{3}{4}}$, so the area is approximately

$$\left(\frac{a^2+1}{2a}\right)\ln\left(1 + \left(k - \tfrac{3}{4}\right)^{-1}\right),$$

as required. Notice that the minimum's position x_2 cancels completely: only the two maxima matter.

Checks

- Differentiating numerically confirms that $\mathrm{d}y/\mathrm{d}x = 0$ at $x = \ln(n + \frac{1}{2})$ for $n = 0, \ldots, 7$ and nowhere else on $[-6, \ln 8]$, and that the gradient at $x = 0$ is $-\pi \ln a$ (for $a = 2$, both are -2.1776).
- With $a = 2$, $k = 1$ the formula gives $\frac{5}{4}\ln 5 \approx 2.012$, while Simpson's rule gives the true area as 1.954: close, as an approximation should be. The trapezia overestimate slightly: the curve dips below the chords around the minimum by more than it bulges above them near the maxima.
- As $k \to \infty$ the area tends to 0, matching the ever-narrower oscillations.

Common slips

- Including negative odd multiples of $\frac{\pi}{2}$ in part (i): $\pi\mathrm{e}^x$ is always positive.
- Numbering the maxima from $k = 0$ rather than $k = 1$, which shifts the answer to $\ln\left(1 + (k + \frac{1}{4})^{-1}\right)$.
- In part (ii), stopping at $a^{-\pi x}$ without the final linearisation $\mathrm{e}^t \approx 1 + t$.

Beyond STEP

Part (ii) is a first-order Taylor expansion done by composing simpler expansions, which is exactly how undergraduates handle expansions of composite functions. The true area over one oscillation, for large k, is about $(x_3 - x_1)$ times the average of $a^{\sin\theta}$ over a period, an integral that leads to the modified Bessel function $I_0(\ln a)$; comparing it with $\frac{1}{2}(a + \frac{1}{a})$ is a nice exercise in why the trapezium rule overestimates for curves like this. For a careful treatment of approximation and Taylor's theorem, see Abbott, *Understanding Analysis*.

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